

Module 05: Action — Renormalization Group and Scale-Free Active Inference

Course: Active Inference 401 | **Unit:** Advanced Theory | **Audience:** Graduate students / researchers

Learning Objectives

1. Analyze the **renormalization group** (RG) as a mathematical framework for multi-scale description.
2. Evaluate how Active Inference can be **scale-free** — applying the same principles from molecules to societies.
3. Examine the formal connections between RG **coarse-graining** and hierarchical model reduction.

Key Concepts

1. Renormalization Group Basics

The renormalization group is a mathematical technique from statistical physics for relating descriptions of a system at different scales:

Coarse-graining: Given a system with many microscopic degrees of freedom (e.g., individual neurons), RG systematically integrates out fine-grained details to produce an effective description at a coarser scale (e.g., cortical columns, brain regions).

Block transformation: Group microscopic variables into blocks, compute effective interactions between blocks, and iterate. At each step: N microscopic variables $\rightarrow N/b$ block variables (b = blocking factor).

Fixed points: Under repeated RG transformations, the effective description flows toward "fixed points" — scale-invariant descriptions that look the same regardless of the observation

scale. Systems at fixed points exhibit critical phenomena (power laws, long-range correlations).

2. Scale-Free Active Inference

Active Inference claims to apply at every scale of biological organization. The RG provides the mathematical framework for this claim:

The same equations at every level: If the free energy functional has the same mathematical form at every RG scale, then the same inference equations apply whether the "agent" is a cell, an organ, a brain, or a society. The specific parameters change (different A, B, C, D matrices), but the computational structure is invariant.

Markov blankets at every scale:

- Molecular: Cell membrane as Markov blanket (receptors = sensory, transporters = active)
- Cellular: Tissue boundary as Markov blanket
- Organismal: Skin/sensory organs as Markov blanket
- Social: Communication channels as Markov blanket

Nesting: Lower-scale blankets are nested within higher-scale blankets. A neuron has its own blanket (cell membrane) nested within the brain's blanket (sensory/motor interface).

3. Coarse-Graining and Hierarchical Reduction

The RG coarse-graining procedure has a direct analogue in Active Inference:

Effective generative models: When fine-grained details are integrated out, the result is an *effective* generative model at the coarser scale. This model has fewer parameters but captures the statistically relevant features of the lower-scale dynamics.

Bayesian Model Reduction at scale: BMR (removing unnecessary parameters) is the inferential analogue of RG coarse-graining (removing unnecessary degrees of freedom). Both simplify the description while preserving essential structure.

Sufficient statistics: At each scale, the effective description is captured by sufficient statistics — the minimal set of quantities needed to characterize the distribution. RG identifies these scaling-relevant statistics.

4. Critical Phenomena and the Brain

The brain may operate near a critical point — a scale-free RG fixed point:

Evidence for criticality: Neural avalanches follow power-law distributions. Long-range temporal correlations exist in neural activity. The brain exhibits "1/f noise" — a hallmark of scale-free dynamics.

Why criticality is useful: Systems at criticality exhibit maximal sensitivity to perturbations (large susceptibility), maximal information transmission (correlation length diverges), and maximal computational capacity. These are exactly the properties an optimal inference machine should have.

Criticality and precision: Operating near criticality may implement optimal precision allocation — the system is maximally responsive to small signal changes (high precision where it matters).

5. From Physics to Society

The scale-free property of Active Inference generates predictions at every organizational level:

Cells: Individual cells minimize free energy by maintaining homeostasis (interoceptive inference) and adapting to their local environment (exteroceptive inference).

Organs: Organs (e.g., the immune system) perform tissue-level inference — identifying pathogens (prediction errors relative to "self" model) and mounting responses (active inference on the immune blanket).

Organisms: Brains perform organism-level inference as described throughout this course.

Societies: Groups of communicating agents form "super-organisms" with their own Markov blankets (communication channels). Cultural norms, institutions, and shared narratives function as shared generative models. Social change = collective model updating.

Summary

The renormalization group provides the mathematical framework for understanding how Active Inference operates at multiple scales. Coarse-graining systematically relates microscopic to macroscopic descriptions. The brain may operate near a critical RG fixed point, maximizing computational capacity. The same free energy minimization principles apply from molecules to societies through nested Markov blankets.

Further Reading

- Friston, K. J. (2019). A free energy principle for a particular physics. *arXiv preprint* arXiv:1906.10184.
- Ramstead, M. J. D. et al. (2018). Answering Schrödinger's question. *Physics of Life Reviews*, 24, 1-16.
- Beggs, J. M. & Plenz, D. (2003). Neuronal avalanches in neocortical circuits. *Journal of Neuroscience*, 23(35), 11167-11177.